GRE Quantitative Reasoning

모의고사 Set 1 — 문제지

시범 세트 · S2 상향

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 1 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Section 1 12 Questions · 21 Minutes

Question 1 대소비교

836. $|2y-5|<1$
Quantity A: y
Quantity B: 1

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

1067. A certain type of code is a list of 4 identical asterisks and 2 identical dots in any order. For example, **•*•* is one such code.
Quantity A: The number of possible codes of this type
Quantity B: 15

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

280. n is the greatest integer for which 252^5 is divisible by 6^n.
Quantity A: n
Quantity B: 10

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

228. On a certain map, 1 centimeter represents 5 kilometers. On the map, region X has an area of 6.4 square centimeters.
Quantity A: The actual area of region X
Quantity B: 150 square kilometers

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

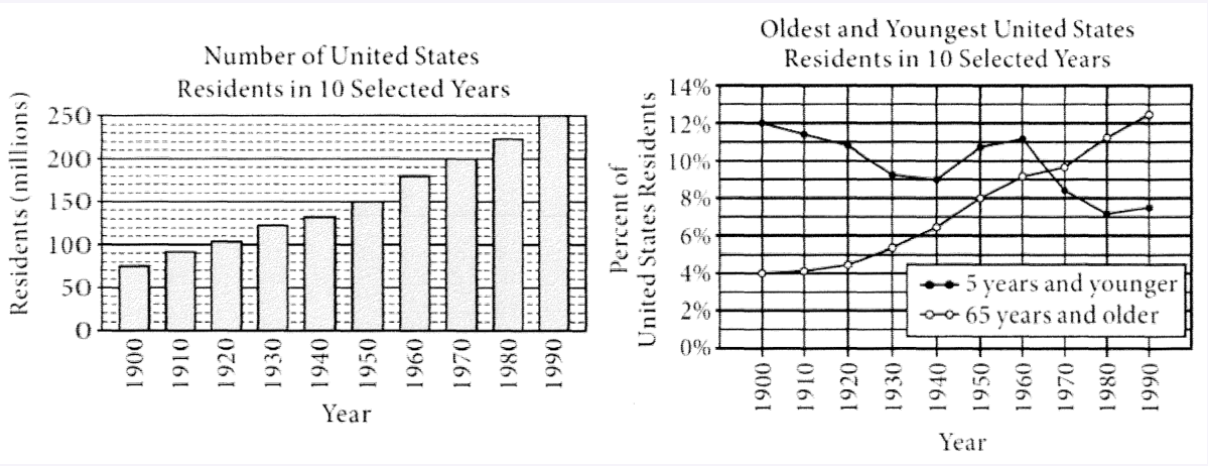
(D) The relationship cannot be determined from the information given.

Question 5 객관식

867. x is an odd integer, and y is a multiple of x. Which of the following must be an even integer?
A xy
B x+y
C x^2y^2
D(x+1)y
Ex(y+1)

(A) (B) (C) (D) (E)

Questions 6 to 8 are based on the following data.



Question 6 객관식

1. In 1970 approximately what percent of United States residents were older than 5 but younger than 65 years old?

A) 2%
B)18%
C) 64%
D) 82%
E) 98%

(A) (B) (C) (D) (E)

Question 7 객관식

2. Of the 10 selected years, which had the greatest total number of United States residents in the combined age groups of "5 years and younger" and "65 years and older"?

A)1900
B)1920
C)1960
D)1980
E)1990

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Question 8 객관식

3. Of the following which best describes how the number of United States residents that were in the age-group "5 years and younger" changed from 1900 to 1940?

A) It increased by more than 50 percent.
B) It increased by less than 50 percent but more than 5 percent.
C) It changed by less than 5 percent.
D) It decreased by less than 50 percent but more than 5 percent.
E) It decreased by more than 50 percent.

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Question 9 객관식

1035. If y=(x+w)/2, what is w−y in terms of x and y?
A3y−x
B2y−x
Cy+x
Dy−x
Ex−y

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Question 10 주관식

754. Passwords for a certain computer consist of 5 symbols typed on a computer keyboard. Each password consists of one @ symbol, two # symbols, and two $ symbols, typed in any order. For example, @#$$# and $#$#@ are two different passwords for the computer. What is the total number of different passwords for the computer?

Question 11 복수선택

359. Sample P: 0.8, 0.9, 0.9, 1.0
Sample Q: 1.0, 1.0, 1.0, 1.0
Sample R: 0.8, 0.8, 1.0, 1.0
Sample S: 0.7, 0.7, 0.9, 0.9
Sample T: 0.9, 1.0, 1.0, 1.1
Five samples of 4 washers each were taken from a production line. The inside diameter of each washer was recorded to the nearest 0.1 centimeter, as shown above. For which of the samples is the standard deviation of the measurements greater than the standard deviation of the measurements for sample P?
Indicate all such samples.
ASet Q
BSet R
CSet S
DSet T

Indicate all that apply. A B C D

Question 12 객관식

1197. There are 10 packages of candy on a shelf, 4 that are filled with blue candies and 6 that are filled with red candies. If 3 packages are to be chosen at random from the shelf without replacement, what is the probability that at least one of the 3 packages will be filled with blue candies?
A. 11/12
B. 5/6
C. 3/4
D. 2/3
E. 7/12

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

678. n is a positive integer
Quantity A: The average (arithmetic mean) of $2^n, 2^{n+1}, 2^{n+2}, 2^{n+3}, 2^{n+4}$
Quantity B: The median of $2^n, 2^{n+1}, 2^{n+2}, 2^{n+3}, 2^{n+4}$

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

164. $a < 0$, $b < 0$
Quantity A: In the xy -plane, the slope of the line containing the points (a, b) and (0, 2)
Quantity B: In the xy -plane, the slope of the line containing the points (a, b) and (2, 0)

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

502. m is an odd number and greater than 1
Quantity A: The greatest prime factor of 2m
Quantity B: The greatest prime factor of m^2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1083. In a company, 1/3 of employees are female, 1/2 of employees have college degrees.
Quantity A: The percentage of female employees who also have college degrees among all employees
Quantity B: 1/6

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 대소비교

679. Three lists contain 17, 35, and 41 measurements, respectively. The median of the measurements in each of the three lists is 52.
Quantity A: The median of the 93 measurements in the three lists combined
Quantity B: 52

(A) Quantity A is greater.

(B) Quantity B is greater.

(C) The two quantities are equal.

(D) The relationship cannot be determined from the information given.

Question 6 객관식

770. For all numbers a and b, the operation **O** is defined by a **O** b=2a-3b. What is the value of 1 **O** (2 **O** 4)?
A-22
B-8
C7
D8
E26

(A) (B) (C) (D) (E)

Question 7 객관식

1018.

$$F = \frac{Gm_1m_2}{d^2}$$

According to a theory, the gravitational force F between two metal spheres with masses m1 and m2, is given by the formula shown, where d is the distance between the centers of the spheres and G is a positive constant. Which of the following changes to the distance or the masses, while keeping the other quantities unchanged, would cause the greatest increase in F?
ADividing the distance by 3
BMultiplying the distance by 4
CMultiplying one of the masses by 3
DDividing one of the masses by 4
EMultiplying each of the masses by 2

(A) (B) (C) (D) (E)

Question 8 객관식

1294. A large cube and a small cube are made of granite of uniform density. The large cube has edges of length 1.2 meters and weighs 4,752 kilograms. If the small cube has edges of length 10 centimeters, what is the weight, in grams, of the small cube?
A.275
B.396
C.570
D.2,750
E.3,960

(A) (B) (C) (D) (E)

Question 9 복수선택

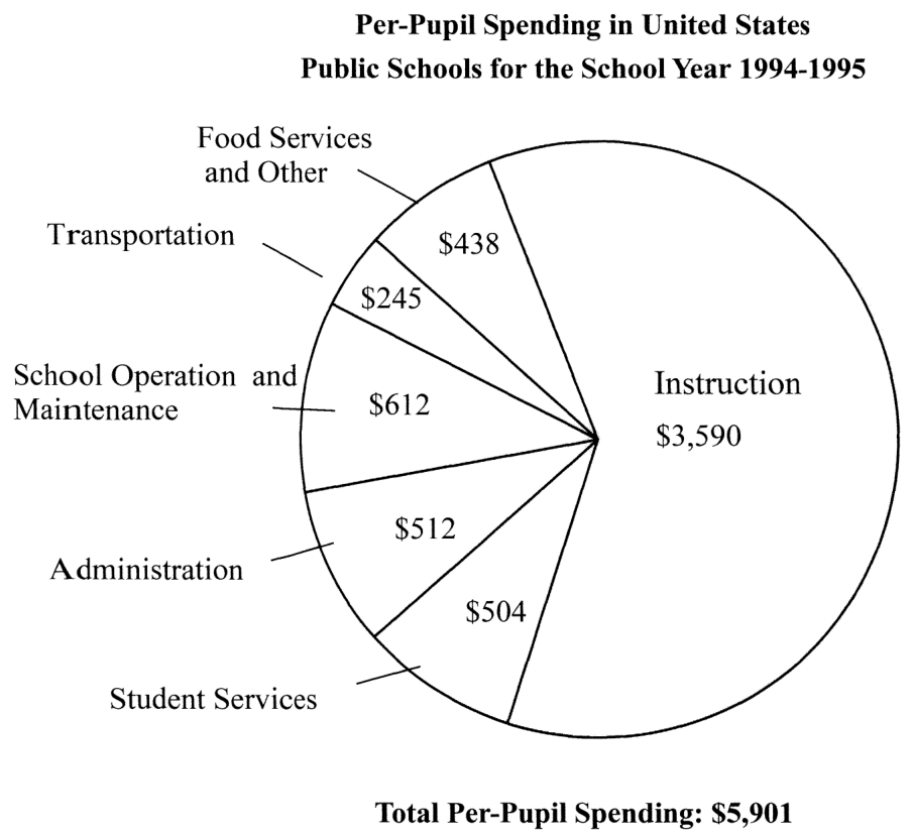
926. Eugene and Penny started a job in sales on the same day. Eugene's sales for the first month were r dollars and each month after the first his sales for that month were twice his sales for the preceding month. Penny's sales for the first month were 10r dollars, and each month after the first her sales for that month were 10r dollars more than her sales for the preceding month. Which of the following statements are true? Indicate all such statements.
AThe dollar amount of Penny's sales for the second month was 10 times that of Eugene's sales for that month.
BThe dollar amount of Penny's sales for the fourth month was 5 times that of Eugene's sales for that month.
CThe dollar amount of Eugene's sales for the eighth month was greater than that of Penny's sales for that month.

Indicate all that apply. A B C

Question 10 주관식

405. A certain museum contains a total of 1,500 works of art, of which 800 are paintings. Of the twentieth-century works of art in the museum, 40 percent are paintings. Of the works of art that are not paintings, 490 are not twentieth-century works of art. How many twentieth-century works of art does the museum contain?

Questions 11 to 13 are based on the following data.



Question 11 객관식

7. Which of the following categories accounted for an amount closest to 10 percent of the total per-pupil spending?

A. Administration
B. Food Services and Other
C. School Operation and Maintenance
D. Student Services
E. Transportation

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Question 12 객관식

8. If the total per-pupil spending for Transportation and Food Services and Other was 10 percent greater for the school year 1994-1995 than for the school year 1993-1994, approximately what was the total per-pupil spending for these two categories for the school year 1993-1994?

A. $620
B. $610
C. $600
D. $590
E. $580

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Question 13 객관식

9. The sectors of the pie chart are to be reordered so that their sizes decrease when viewed consecutively in a clockwise direction starting with the largest. If the sectors are to be reordered by interchanging the positions of adjacent sectors, one interchange at a time, until the desired order is achieved, what is the minimum number of interchanges that will be required?

A. Three
B. Four
C. Five
D. Six
E. Seven

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Question 14 객관식

431. Four people, including Ito and Kenji, are to be assigned to 3 offices. Two of the offices can accommodate only 1 person and the other office can accommodate 2 persons. If Ito and Kenji cannot be assigned to the same office,how many different assignments of the 4 people to the offices are possible?
ASix
BSeven
CEight
DTen
EEleven

A B C D E

Question 15 복수선택

893. In 2009, the price of a certain stock increased by 25 percent from May 1 to June 1 and decreased by n percent from June 1 to July 1. If the price of the stock on July 1 was greater than the price of the stock on May 1, which of the following could be the value of n? Indicate all such values.
A13
B18
C23
D28

Indicate all that apply. A B C D